

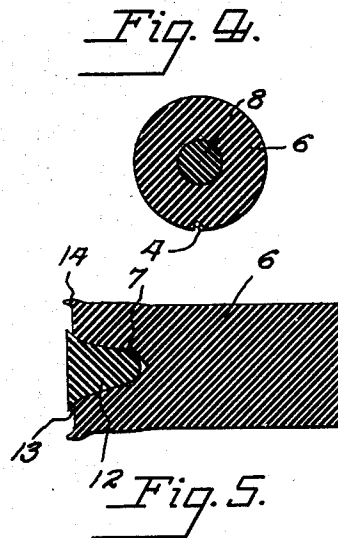
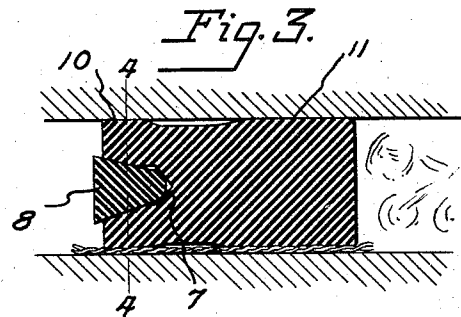
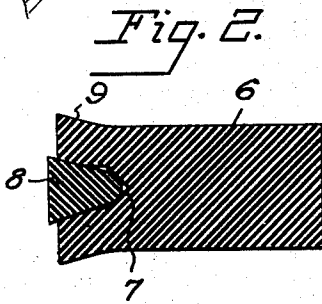
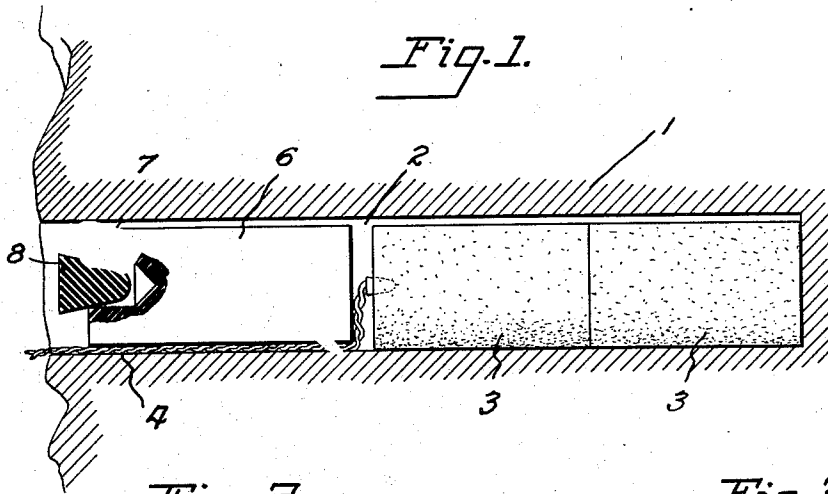
July 9, 1935.

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2,007,568

BLASTING PLUG

Filed Dec. 6, 1934



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UNITED STATES PATENT OFFICE

2,007,568

BLASTING PLUG

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Application December 6, 1934, Serial No. 756,319

2 Claims. (Cl. 102—11)

This invention relates to blasting plugs and it is an improvement upon the blasting plugs illustrated and described in my copending application for patent, Serial No. 747,435, filed October 8, 1934.

One of the objects of the invention is to provide a plug of elastic material such as rubber, having a tapered insert seating in a recess in the outer end and adapted to be forced into the recess after the plug is in place in the drilled hole in the rock or coal, for expanding the outer end of the plug compressively against the surrounding wall of the drilled hole, making a gas-tight seal and immovably fixing the plug in position.

Another object of the invention is to provide a blasting plug that shall offer no incentive for its removal in case of misfire, thus eliminating the dangerous practice of drilling out the plug.

A further object of the invention is to provide a blasting plug that shall obviate the need of "stemming" the bore in the coal or rock.

Still another object of the invention is the provision of a blasting plug so constructed as to cause the explosion to bring down an increased amount of coal or rock and produce an increase in the percentage of large pieces.

Other objects of the invention will appear as the following description of a preferred and practical embodiment thereof proceeds.

In the drawing which accompanies and forms a part of the following specification, and throughout the several figures of which the same characters of reference have been employed to designate identical parts:

Figure 1 is a longitudinal section of a hole drilled in the coal or rock showing the explosive in place and a blasting plug embodying the principles of the present invention in place, but not, as yet, in expanded condition;

Figure 2 is a longitudinal section through the blasting plug showing the tapered insert driven home, expanding the periphery of the outer end of the plug;

Figure 3 is a longitudinal section through the plug showing its condition immediately following an explosion;

Figure 4 is a section taken along the line 4—4 of Figure 3 illustrating the manner in which the plug yields to embed the wires; and

Figure 5 is a longitudinal section through a modified form of plug illustrating a tapered insert with flaring base producing a peripheral lip at the outer end of the plug.

Referring now in detail to the several figures

the numeral 1 represents a mass of coal or rock in which is drilled the bore or hole 2, the remote portion of the hole being filled with the sticks 3 of dynamite or other suitable explosive to which lead the electric wires 4 terminating in a fulminating cap 5 inserted in the end of the dynamite. It is customary in the art to fill the anterior part of the hole with small bags of sand or earth tamped into place, the operation being known as "stemming". The sand or earth being quite abrasive, it frequently happens that the insulation is worn off of one or both of the electric wires grounding them, so that when the circuit is closed, a misfire occurs and no explosion takes place. Since a number of charges are set off simultaneously, it is frequently impossible to ascertain which if any of the charges failed to explode and an unexploded charge may be thrown down to the floor of the mine or quarry by the breaking up of the rock or coal caused by an explosion of adjacent charges. The unexploded dynamite is frequently indistinguishable from the debris and there is the hazard of the miner striking it with his pick and exploding it with the loss of life or limb. In any event, "stemming" is a time-consuming and expensive operation.

The plug 6 of the present invention obviates the need of "stemming", for it is designed to be held in place initially prior to the explosion and further fixed in position by the force of the explosion. With this end in view, the plug is formed with a cylindrical body of rubber or other substance capable of expanding uniformly in a radial direction. In its initial shape it just freely fits the drilled hole so that it can be slipped into place by the mere pressure of the tamper. In its outer end the plug 6 is provided with a well or depression 7 preferably cylindrical in its outer portion or slightly flared. An insert 8 seats in said depression, being larger in diameter than said depression at its outer end and tapering towards its opposite end which is sufficiently small to extend freely within said depression. The insert may be made of rubber or of any other suitable material. It is not necessary that it be yielding since its object is to deform or spread the outer end of the plug 6 when driven into said depression. After the plug 6 has been pushed gently into position against the explosive charge, the tamper is then forcibly pushed or struck against the end of the insert 8 driving it into said depression. The rubber or other explosive material of which the plug is made is radially diverted as indicated at 9 in Figure 2, defining a peripheral

flare which presses tightly against the surrounding wall of the drilled hole making a fluid-tight seal as at 10 in Figure 3, and holding the plug firmly against displacement.

5 It is absolutely essential in order for the explosive to exert its full force against the coal or rock that the explosion chamber be sealed against escape of the gases of explosion, at the moment of explosion, otherwise the intensity of the pressure is greatly lessened. It is due to the perfect seal indicated at 10 that the plug of my invention causes the explosion to bring down a greater amount of coal or rock than would otherwise be the case.

15 On account of its elastic quality and the fact that it is relatively heavy and therefore possesses considerable inertia, the plug retains itself in position despite the intensity of the explosion and therefore requires no "stemming".

20 At the instant of explosion the pressure against the inner end of the plug expands it radially compressing it against the wall of the drilled hole as indicated at 11 throughout an extensive area jamming it in the hole and preventing its being

25 blown out. It will be observed from Figures 3 and 4 that when the anterior end of the plug is expanded by tamping, and later when the inner end is expanded through the force of the explosion, the rubber or other material yields to the presence of the wires 4 as indicated in Figure 4, so that the wires embed themselves in the body of the plug and are safeguarded from abrasion. Initially, if the plug is a tight enough fit in the drilled hole, the wires press against the surface of the plug as it is shoved in, forming a seat in which they are yieldingly embedded.

35 In view of the fact that "stemming" is not necessary and that only a light blow of the tamper is essential to drive the insert 8 into the depression 7, the wires are not subject to abrasion, consequently, a short circuit is not likely to occur and there is practical assurance that the explosion will take place. In the event that a misfire does occur, the explosion may be induced by firing a charge of dynamite against the outer surface of the plug.

40 In view of the yielding nature of the material of which the plug 6 is made, the impact of the explosion is slightly cushioned, so that the force of the explosion is not so much expended in shattering the proximate mass of rock or coal, but is given them for transmission deeper into the mass, throwing down more coal or rock and larger pieces.

55 It is preferred as has been previously stated to make the plug of rubber, although any other material or composition having like characteristics is within the purview of the invention. The selection of material for the insert 8 is not very critical since the relative hardness or softness does not matter providing of course that the insert is firmer than the material of the plug 6

so that it can transmit a deforming pressure to the plug.

Figure 5 shows a slightly modified form of the invention in which the insert 12 differs from the insert 8, previously described, in having a flared peripheral flange 13 at its outer end. When the flange portion of this insert is driven into the depression 7 of the plug, the material of the plug is deformed so as to throw out a relatively thin lip 14 peripherally of the outer end of the plug. This lip extends toward the exterior of the hole and tends to push obliquely against the wall of the drilled hole in a direction to become more tightly engaged with said wall as the force which would displace the plug outwardly increases. This is an added safeguard against the blowing out of the plug under the force of the explosion.

While I have in the above description disclosed what I believe to be a preferred and practical embodiment of my invention, it will be understood by those skilled in the art that the details of construction as herein illustrated and described are merely by way of example and not to be construed as limiting the scope of the invention as claimed.

What I claim is:

1. Blasting plug comprising a body of yielding rubber adapted to be inserted in a drilled hole in coal or rock adjacent a mass of explosive, said body having a recess in its outer end and a tapered insert normally seated in the anterior portion of said recess and adapted to be forced deeper into said recess for expanding the external periphery of the outer end of said plug into compressed relation to the walls of said drilled hole, the inner portion of said body extending beyond any other part of said plug so as to expose said portion direct to the explosion pressure, thereby providing a yielding cushion for mollifying the shattering effect of the explosion, and adapted to be expanded by the pressure of the explosion into sealing relation to the wall of the drilled hole.

2. Blasting plug comprising a non-metallic unit consisting of a body composed wholly of yielding rubber adapted to be inserted in a drilled hole in coal or rock, having a recess in its outer end and a tapered insert normally seated in the anterior portion of said recess, and adapted to be forced deeper into said recess for expanding the external periphery of the outer end of said plug into compressed relation to the walls of the drilled hole, the inner portion of said body extending beyond any other part of said plug so as to expose said portion direct to the explosion pressure, thus providing a cushion for yieldingly resisting the explosion to mollify the shattering effects thereof, and the inner end of said body being adapted to be expanded by the pressure of the explosion into sealing relation to said walls.

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